

OpenClaw AI Pet Tutorial

Before starting, please complete [01-Peripheral Hardware Configuration], [02-OpenClaw API-KEY Configuration], [03-OpenClaw Model Switching], [04-AI Voice Interaction Configuration], and [05-Three Dialogue Configuration Tests] in the `Preparation Before Use` directory. If you are only using TUI/WhatsApp, you can skip `04`. This article assumes the basic environment is ready and only covers the AI pet project itself.

1. Tutorial Objectives

This article demonstrates OpenClaw's interactive capability as a desktop AI pet through the combination of "actions + lights + screen + dialogue". After completing this tutorial, readers will be able to:

- Understand how to create an AI pet with emotional feedback using servo, RGB, and OLED
- Communicate with OpenClaw via WhatsApp, TUI, and voice commands
- Complete four project scenarios: greeting, emotional expression, companionship Q&A, sleep and wake

2. Hardware Preparation

- OpenClaw main control board
- Servo [recommended at least 1 unit]
- RGB light strip / RGB LED module
- OLED display screen
- [Optional] AI voice module + speaker
- Wiring diagram



3. Software Preparation

Code entry points used in this tutorial:

- Servo control command: `/home/pi/hardware_hal_demo/cli/servo`
- Voice entry: `/home/pi/voice_to_openclaw/src/main.py`

This project does not have a fixed set of "pet scripts", but lets OpenClaw orchestrate several basic peripheral actions:

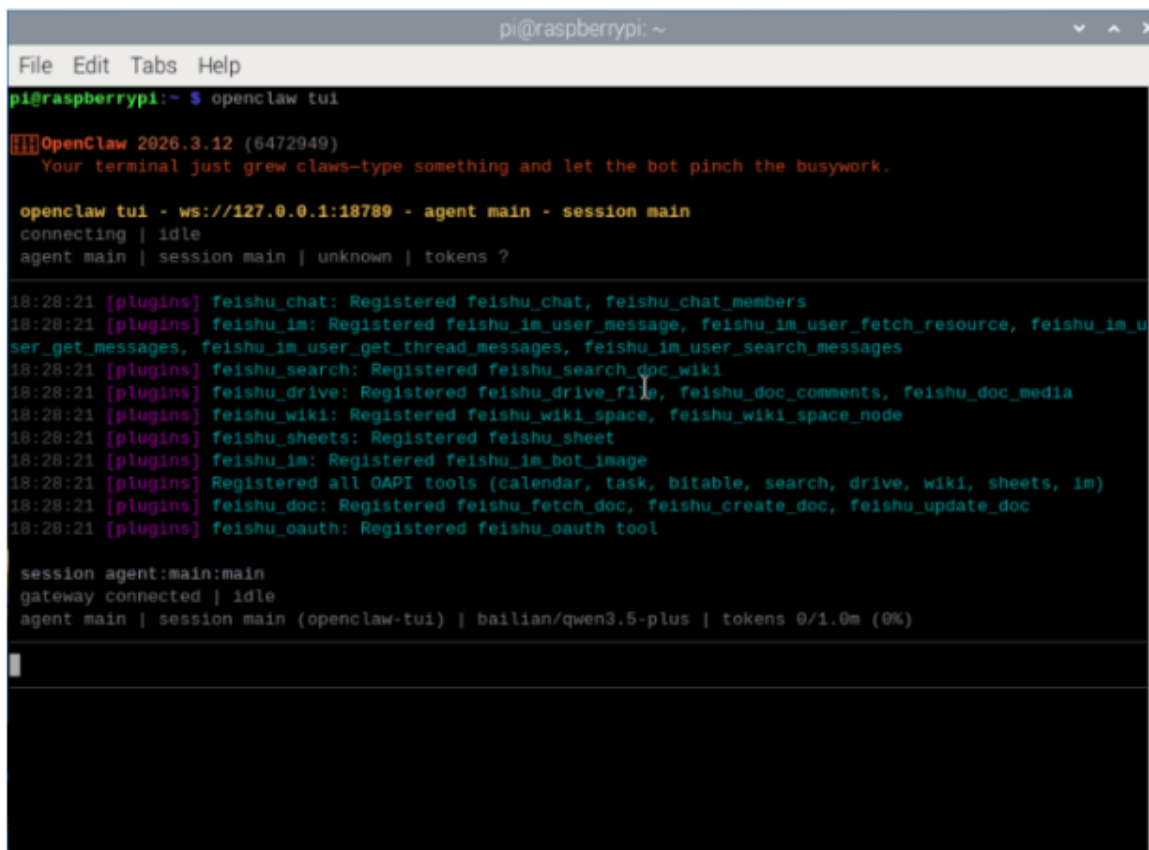
- Servo is responsible for nodding, head shaking, and swinging
- RGB is responsible for expressing mood and status
- OLED is responsible for displaying names, expressions, or short text
- Voice dialogue is responsible for making the pet more like an interactive character

4. Testing Project Basic Capabilities

Before starting this project, please complete an `openclaw tui` basic conversation self-check first; if you haven't done this yet, see [05-Three Dialogue Configuration Tests]. After passing, proceed with the AI pet project-specific test.

Enter OpenClaw TUI:

```
openclaw tui
```



```
pi@raspberrypi: ~
File Edit Tabs Help
pi@raspberrypi:~ $ openclaw tui

[!!!] OpenClaw 2026.3.12 (6472949)
  Your terminal just grew claws-type something and let the bot pinch the busywork.

openclaw tui - ws://127.0.0.1:18789 - agent main - session main
connecting | idle
agent main | session main | unknown | tokens ?

18:28:21 [plugins] feishu_chat: Registered feishu_chat, feishu_chat_members
18:28:21 [plugins] feishu_im: Registered feishu_im_user_message, feishu_im_user_fetch_resource, feishu_im_u
ser_get_messages, feishu_im_user_get_thread_messages, feishu_im_user_search_messages
18:28:21 [plugins] feishu_search: Registered feishu_search_doc_wiki
18:28:21 [plugins] feishu_drive: Registered feishu_drive_file, feishu_doc_comments, feishu_doc_media
18:28:21 [plugins] feishu_wiki: Registered feishu_wiki_space, feishu_wiki_space_node
18:28:21 [plugins] feishu_sheets: Registered feishu_sheet
18:28:21 [plugins] feishu_im: Registered feishu_im_bot_image
18:28:21 [plugins] Registered all OAPI tools (calendar, task, bitable, search, drive, wiki, sheets, im)
18:28:21 [plugins] feishu_doc: Registered feishu_fetch_doc, feishu_create_doc, feishu_update_doc
18:28:21 [plugins] feishu_oauth: Registered feishu_oauth tool

session agent:main:main
gateway connected | idle
agent main | session main (openclaw-tui) | baillan/qwen3.5-plus | tokens 0/1.0m (0%)
```

Then separately test several basic capabilities:

- Make the servo nod
- Change the lights to pink breathing light
- Display "Hello" on OLED

5. Three Ways to Communicate with OpenClaw

Method	Suitable Scenario	Advantages	Suggested Use
WhatsApp	Remote interaction, classroom demonstration	Suitable for demonstrating "remote pet keeping"	Demonstrate character interaction capability
TUI	Local debugging, action verification	Easiest to check if actions and feedback match	First to get character action logic working
Voice Code	Ultimate interactive experience	Most like a real chatty desktop pet	As the final demonstration solution

If you have the optional AI voice module + speaker, the TUI and WhatsApp conversations need to broadcast OpenClaw's replies. You can run the following program in the terminal. This is not needed for AI voice module conversations.

- Modify the TTS enable parameter

```
/home/pi/voice_to_openclaw/.env
```

```
ENABLE_TTS=true # This parameter controls whether to broadcast replies. After saving, run the program below in the terminal.
```

- Terminal input

```
cd ~/voice_to_openclaw
source .env/bin/activate
python -m src.openclaw_session_tts
```

```
(.venv) pi@raspberrypi:~/voice_to_openclaw $ python -m src.openclaw_session_tts
[系统] OpenClaw 会话播报器已启动
[系统] 监听目录=/home/pi/.openclaw/agents/main/sessions
[系统] 监听渠道=feishu,tui,whatsapp
```

5.1 WhatsApp Solution

For WhatsApp integration and bot creation steps, please refer directly to [05-Three Dialogue Configuration Tests]. After creating, you can interact with the AI pet remotely in WhatsApp, for example:

- Take a photo to see if there is dog food. If yes, display smile expression on OLED, make servo dance, and light up ambient lights on OLED. If no dog food, display sad expression on OLED.

20:31



< 2



+86 153 3885 7526 (你)

给自己发消息

Take a photo to see if there is dog food. If yes, display smile expression on OLED, make servo dance, and light up ambient lights on OLED. If no dog food, display sad expression on OLED.

20:31 ✓✓



I can see the scene, but there's **no dog food** visible in the image. I can see a desk with various items like a pink mug, green fan, water bottle, and a curriculum schedule, but no dog food detected

20:31 ✓✓

Dog food check completed!

Result: No dog food detected in the scene



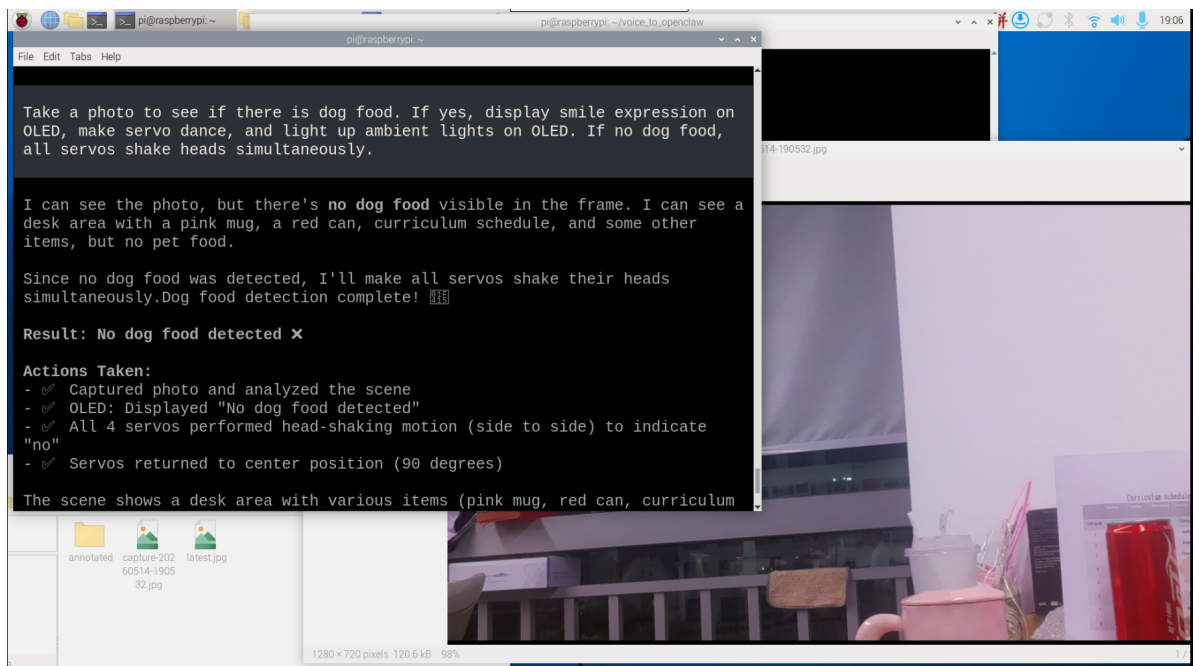
5.2 TUI Solution

If you have already completed the TUI self-check from chapter 0, directly enter terminal text dialogue:

```
openc1aw tui
```

Then input project-related commands, for example:

- Take a photo to see if there is dog food. If yes, display smile expression on OLED, make servo dance, and light up ambient lights on OLED. If no dog food, all servos shake heads simultaneously.



5.3 Voice Code Solution

Requires optional AI voice interaction module + speaker

For voice module installation, environment configuration, and basic wake-up flow, please refer directly to [04-AI Voice Interaction Configuration]. After waking up, you can directly speak control content, for example:

- Take a photo to see if there is dog food. If yes, display smile expression on OLED. If no dog food, display sad expression on OLED.


```
[你] Take a photo to see if there is dog food if yes display smile expression on  
old if no dog food display sad expression.  
[发送] 正在交给 OpenClaw  
[完成] OpenClaw 返回 200  
  
[OpenClaw] ## 📷 Dog Food Detection Complete!  
  
**Result:** ✖ **No dog food  
found**  
  
**What I see in the image:**  
- Window with gray roller  
blind  
- Pink mug and red Coca-Cola can  
- Curriculum schedule paper  
-  
Some boxes and containers on shelves  
- Tissue box  
  
**No pet food or  
dog food detected!** 😞  
  
**Feedback:**  
- OLED displaying sad face:  
`:(`  
- RGB lights set to dim purple (sad mood)  
  
Want me to check  
again later or look for something else? 📷  
[信息] 回复已裁剪为播报文本: 437 -> 121 字
```

6. Project Implementation

Note: All demonstrations below use the `openclaw tui` method for conversation. You can also choose WhatsApp or voice to complete.

6.1 Scenario 1: Greeting

Experiment Objective

Have the AI pet respond with actions and text feedback when the user sends a greeting.

Effect Description

The servo can do a simple nodding or head-shaking action, OLED displays welcome text, and RGB lights up with soft colors, so the feeling of the pet "appearing" will be very obvious.

Example Commands

- Have the little pet say hello to me
- Have the pet welcome me

6.2 Scenario 2: Emotional Expression

Experiment Objective

Have the AI pet express states such as happy, quiet, angry, and sleepy using lights and screen.

Effect Description

This scenario is most suitable for reflecting "character feeling". For example, warm-colored breathing light when happy, blue glow when sleepy, and only short status when quiet.

Example Commands

- Have the pet enter happy state
- Draw simple expressions on OLED

6.3 Scenario 3: Companionship Q&A

Experiment Objective

Have the AI pet, in addition to being able to move, respond to questions like a desktop companion character.

Effect Description

OpenClaw can first complete the dialogue, then combine with servo nodding, head shaking, and screen text to make "speaking" and "actions" happen simultaneously, making it look more like a character with personality.

Example Commands

- First answer my question, then nod
- If you agree, express it with actions
- Chat with me about today's mood

6.4 Scenario 4: Sleep and Wake

Experiment Objective

Have the AI pet have the ability to switch between standby, sleep, and wake states.

Effect Description

In sleep mode, most lights can be turned off, keeping only low-brightness status; when waking up, restore lights, screen, and action feedback, making the project more complete.

Example Commands

- Have the pet enter sleep mode
- Wake up the pet